Student Performance Predictor – Project Report

1. Problem Definition

This project aims to build a machine learning model that predicts whether a student will pass or fail based on specific features. The problem is a binary classification task where the target variable is labeled as 'pass' or 'fail'.

2. Data Source and Preprocessing

The dataset used for this project is 'student-mat.csv', originally from the UCI Machine Learning Repository. It contains student information from a Portuguese secondary school. Preprocessing steps include:
- Removing missing values
- Label encoding categorical features
- Creating a binary target variable from the 'G3' grade
- Selecting 8 key numeric features for the model: age, studytime, failures, absences, goout, Dalc, Walc, health

3. Model Description and Performance Metrics

The model used is a Random Forest Classifier with 100 estimators. It was trained using 80% of the dataset and tested on the remaining 20%. Evaluation metrics:
- Accuracy: around 0.84 (exact may vary based on random seed)
- Classification Report: includes precision, recall, and F1-score for each class
- Confusion Matrix: helps visualize true/false positives and negatives

4. Screenshots of Flask Application

Screenshot of the Flask web interface can be inserted here manually in the final document.

5. Reflection

What Worked:
- Model training with key features produced good results
- Flask app allows user-friendly predictions
- Form-based interface ensures easy input

What Didn't:
- Original model used too many features; had to retrain for compatibility with web UI

- Ensuring input feature alignment required careful restructuring

Future Improvements:
- Add more features to improve accuracy
- Implement model explainability (e.g., SHAP values)
- Improve UI with dropdowns and data validation
- Deploy online using services like Heroku or Render